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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

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1. Team & Project Overview

Team Name	E30S
Selected Domain	DOMAIN 2
Selected SDG Track (1–17)	SDG 1, 4, 8, 9, 12, 16
Project / Solution Name	BISCOACH
Name (Leader)	MUHAMMAD HAIRIL ZAFRAN BIN MOHD HELMI
Email & Phone (Leader)	0102280047 / hairilzafran4@gmail.com
University Affiliation(s)	UNIVERSITI MALAYA
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small and medium-sized enterprise (SME) owners, especially in food and retail, run daily operations on disconnected tools. Transactions are scattered across paper receipts, handwritten notes and separate spreadsheets, while inventory and invoices are tracked elsewhere. Staff lose hours re-keying entries and hunting for missing transaction details.

This matters because incomplete, non-real-time records make it hard for owners to see true profit, track expenses or make confident daily decisions. Without a connected view of business performance, records stay inconsistent and errors go unnoticed.

The impact reaches beyond the business itself:

Economic: Manual processes cause high error rates and revenue losses. Poor stock visibility locks up capital and raises carrying costs, and inaccurate demand forecasting leads to over- or under-ordering.

Environmental: Over-ordering and weak tracking accelerate food spoilage, which produces greenhouse gas emissions and wastes the water, land and energy used upstream to produce the food. Excess stock also adds to packaging waste.

Social: Wasted edible food worsens food insecurity. Repetitive administrative work causes fatigue and burnout among employees. Poor records also create public health risks and regulatory non-compliance, for example when expiry dates and batches go untracked.

By bringing transactions, inventory and invoicing into one system, our solution gives SME owners reliable real-time data to cut waste, protect margins and operate responsibly.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Most Malaysian micro and small businesses have one to five people, no HR system and no accountant on staff. The owner is also the stock clerk, bookkeeper and marketer. BISCOACH is a mobile-first web app built for this owner-operator: an AI business coach that handles the daily admin that employee-focused platforms never reach, namely stock, supplier costs, receipts and marketing.

How AI solves the problem:

- **Conversational agent:** Owners type in English, Bahasa Malaysia or Manglish (for instance, ‘jual 3 air milo’). Using tool-calling, the agent searches inventory and drafts stock and transaction changes, so nobody fills in a spreadsheet.
- **Receipt scanner:** Gemini converts a photo or PDF into structured fields (merchant, items, tax, total, category). Food products also carry expiry dates, helping owners prevent waste.
- **Vendor Overcharge Checker:** Gemini only reads invoices and contracts. Deterministic code then flags math errors, price breaches and quantity gaps, protecting margins with explainable results.
- **Smart posting schedule:** a bandit-style recommender, not an LLM, learns the best posting days, and Gemini drafts captions.

The AI only proposes. Every change needs the user’s approval, and model output is schema-validated in code, so owners stay in control.

Skills from SDC Phase 1

We apply the technical skills we gained using our own stack:

- GenAI API integration: structured output, tool-calling and prompt design.
- Database architecture: multi-tenant data modelling and real-time sync.
- Cloud hosting: server-side API design and deployment.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Technologies for the deployed prototype:

- **Frontend and backend:** Next.js (React), with server-side API routes for all AI calls, so API keys never reach the browser
- **Database:** Firebase Firestore for transactions, inventory, social media view logs, support tickets, compliance checklist and chat history
- **Authentication:** Firebase Authentication for owner and staff login
- **Cloud hosting (SDC sponsor-provided):** Replit Deployments, with API keys kept in Replit Secrets
- **GenAI APIs:** Google AI Studio Gemini API (Gemini 3.5 Flash, with Gemini 2.5 Flash as fallback) for the chatbot, receipt processing, text-to-inventory, vendor document reading and video ideas
- **Receipt images:** processed in memory and not stored

We used the SDC sponsor-provided cloud platform (**Replit**) to host and publish the prototype. We did not use any sponsor-provided GenAI API: AI features run on the Google AI Studio API. Our database and authentication use Firebase.

"I attach the diagram at the appendix section for the system block diagram"

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Design-Thinking Framework & Empirical Validation

- **Empathise.** We started from the owner-operator's day: one person handling stock, receipts, suppliers and marketing. This shaped a mobile-first design and input in English, Bahasa Malaysia and Manglish.
- **Define.** The core problem is manual record-keeping and costly supplier errors, not a lack of dashboards.
- **Ideate.** We chose AI where it removes typing (chat, receipt scanning), and plain code where trust matters (invoice checks), so results stay explainable.
- **Prototype.** We built a working Next.js app with Firebase, with guided onboarding by business type so new users aren't left facing an empty screen.
- **Test.** Technically, we added fallback models, schema-validated AI output and server-side API calls to protect keys. Nothing is saved until the user approves it.
- **Ready for Phase 2.** The multi-tenant data model supports many businesses, and an API credit monitor tracks usage costs. Next, we pilot with real SMEs, measure time saved and errors caught, and iterate.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	● **1.4** micro-enterprises reach financial aid and credit schemes ● **4.4** the AI coach closes the accounting and digital skills gap ● **8.3** Promotes MSME formalisation and financial access ● **9.3** Increase access of small-scale enterprises to financial services ● **12.3** less food waste ● **12.5** - Reduce waste Generation through inventory tracking ● **12.6** Encourage companies to integrate sustainability reporting. ● **16.6** transparent, accountable records: audit-ready ledger, e-invoice, compliance
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	● 89% lack the skill and expertise to prepare financial statements, citation **[Mohamed et al. (2023), Insight Journal 10(1)]** ● 80.3% of Malaysian businesses use social media (reference year 2024), citation **[DOSM via The Edge, 30 Jun 2026]** ● 57% of firms did not track social media's effect on performance (PwC 2014), citation **[Md Jani et al., USM ART_6]**, which cites PwC secondhand ● Food forgotten in storage until it expires is the biggest cause of food waste, followed by over-buying and overstocking, citation **[The Star, 9 Jul 2026]**

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	• **Financial Record Completeness Rate (%) :** The percentage of a pilot merchant's receipts and transactions captured as complete, structured ledger entries (date, merchant, items, tax, total, category) without needing accounting expertise. • **Posting Consistency and Performance Tracking Rate (%):** the percentage of planned weekly posts published on the recommended schedule, and the percentage of posts with logged performance (views or engagement) per merchant • **Expiry-Related Food Waste (kg per month, % reduction):** The weight (or unit count) of food stock expired unsold per pilot business per month, and the percentage of near-expiry items sold or used after an alert.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	• **Financial Record Completeness**: 90% of receipts captured as complete, structured ledger entries **Target:** at least 90% of pilot merchants' receipts and transactions are captured with all six fields correct (date, merchant, items, tax, total, category), checked on a hand-verified sample of at least 50 receipts, including faded or crumpled ones. • **Posting Consistency and Performance Tracking**: 70% of planned posts published, 80% with logged performance **Target:** at least 70% of planned weekly posts go out on the recommended schedule, and at least 80% of posts have views or engagement logged. • **Expiry-Related Food Waste**: 25% reduction in expired-unsold stock **Target:** a 25% drop in units or kg expired unsold per

Metric Category	Your Response
	pilot business per month, and at least 60% of items flagged as near-expiry sold or used before they expire.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Next.js	Code (framework)	MIT
React	Code (library)	MIT
Firebase JS SDK (firebase/app, auth, firestore)	Code (library)	Apache-2.0
@google/genai (Google Gen AI SDK)	Code (library)	Apache-2.0
@zxing/library and @zxing/browser (barcode scanning)	Code (library)	MIT
Poppins font	Font asset	SIL Open Font License 1.1
Inter font	Font asset	SIL Open Font License 1.1
Gemini 3.5 Flash (fallback: Gemini 2.5 Flash), accessed via API	Pre-trained AI model	Proprietary (Google), used under the Gemini API Terms of Service

6.3 Employment-Related IP Conflicts

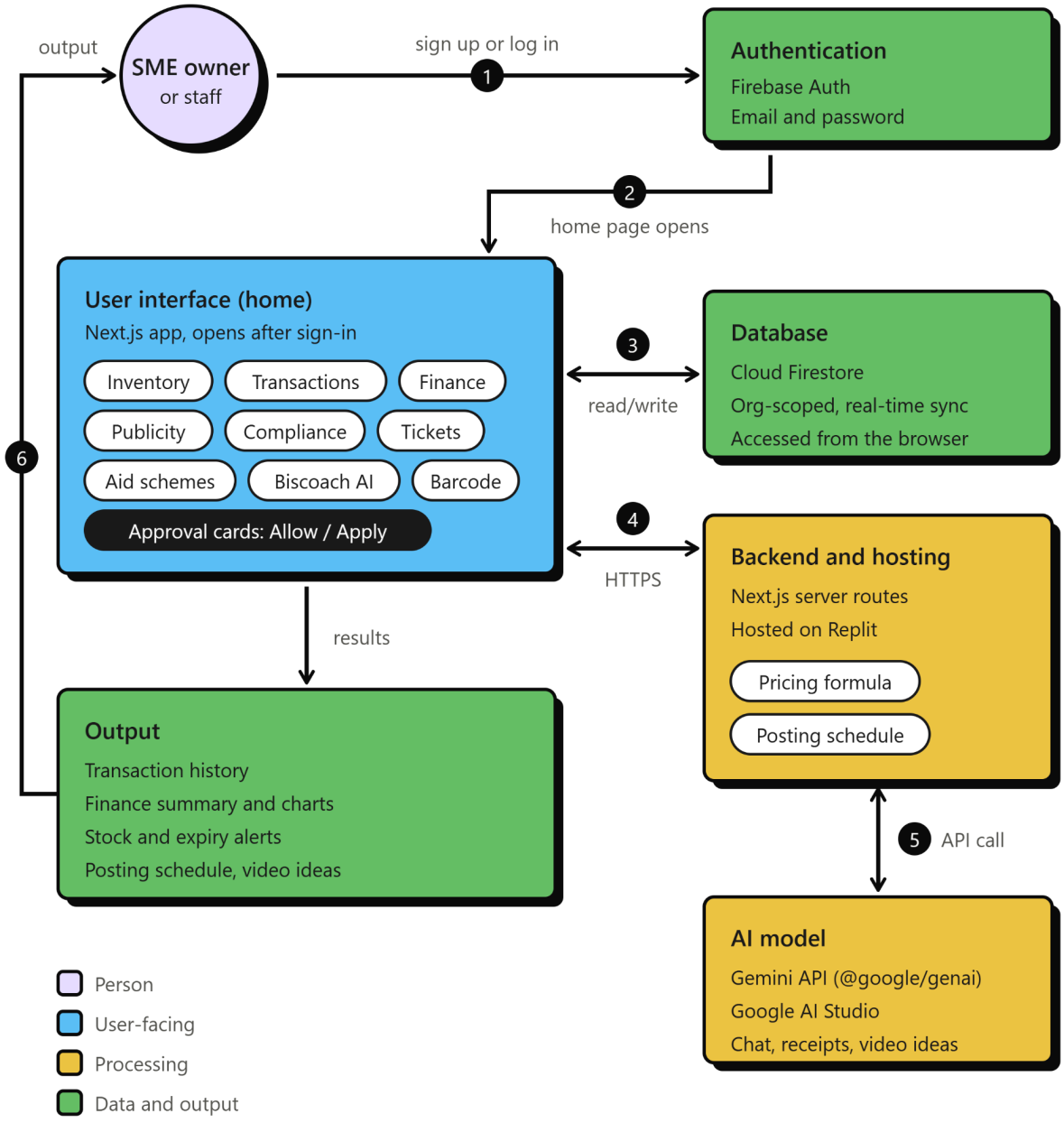
Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Appendix

3.1.



8. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



MUHAMMAD HAIRIL ZAFRAN BIN MOHD HELMI
24 OCTOBER 2026